

## INVESTIGATION REPORT

# Issue 333: Reading a debug-only CSR outside debug mode must trap (closed bug, replayed)

openhwgroup/cv32e40x

|                       |                                                         |
|-----------------------|---------------------------------------------------------|
| <b>Document</b>       | RTL-TRIAGE-CV32E40X-333                                 |
| <b>Disposition</b>    | Bug confirmed, fix verified                             |
| <b>Record date</b>    | 2026-09-11                                              |
| <b>Classification</b> | Internal working document. Not a verification sign-off. |

## 1 Status

|                      |                                                      |
|----------------------|------------------------------------------------------|
| <b>Issue</b>         | openhwgroup/cv32e40x#333                             |
| <b>Issue type</b>    | Bug report from an integrator or user                |
| <b>Affects</b>       | cv32e40x                                             |
| <b>Configuration</b> | pre-fix RTL 4ac00cd2 (2021-11-25) vs master; X_EXT=0 |
| <b>Disposition</b>   | Bug confirmed, fix verified                          |
| <b>Severity</b>      | Not assessed                                         |
| <b>Agreement</b>     | Same disposition as the independent human record.    |
| <b>Evidence</b>      | 6 cited, 2 present, 4 not retained                   |

*Disposition meaning. The reported behaviour was reproduced, and the checks listed under Evidence pass against the corrected design.*

*Severity is not recorded for this run. The harness does not infer a severity from the disposition, so the field is reported as not assessed rather than filled in.*

## 2 Description

Confirmed on the pre-fix RTL and verified fixed on master. Property (controller\_fsm scope): a CSR instruction (opcode 0x73, funct3 != 0) addressing dcsr/dpc/dscratch0/dscratch1 (0x7B0-0x7B3) that retires in WB while not in debug mode must raise an illegal-instruction exception, unless the core is entering debug on that instruction (trigger match / DEBUG\_TAKEN). Environment: no instruction/data bus errors. On RTL 4ac00cd2 (2021-11-25) BMC finds an 8-cycle counterexample: csrrs x0, dscratch0, x0 (0x7b202073) in M-mode; cs\_registers flags csr\_illegal\_o=1 in EX at cycle 6, yet the instruction retires in WB at cycle 7 with no exception (no debug, no trigger, no interrupt, no bus error). On master the property is proven for all time (abc pdr) and the covers show the same instruction traps (c\_hit\_traps at cycle 7) while the same access in debug mode is legal (c\_dbg\_ok). Five property iterations were needed, each diagnosed from a trace: v1 used the pipeline csr\_en flag (vacuous: the flag is cleared for illegal ops), v2 used the instruction word, v3 added quiet-bus assumptions (solver had injected an instruction-bus error, cause 1 instead of 2), v4 required wb\_valid and excluded ctrl\_fsm\_ns==DEBUG\_TAKEN, v5 also excluded ctrl\_fsm\_cs==DEBUG\_TAKEN and trigger\_match\_in\_wb (solver armed a trigger on the CSR instruction's address, a legal debug entry that suppresses the exception). Comparison with the maintainers' fix, read afterwards (commit a34730a6, 2021-12-01, PR #335 'Fixed issues #332 and #333, where illegal CSR instructions were not properly handled'): in the EX stage the qualifier was csr\_illegal\_i && (xif\_en && !xif\_accepted), which with X\_EXT=0 is identically false, so the illegal flag never reached WB; the fix inverts it to csr\_illegal\_i && !(xif\_en && xif\_accepted). That is exactly the gap the counterexample shows (flag set in EX, no exception in WB).

## 3 Conditions

*Not recorded as separate fields for this run. The conditions and environment assumptions that bound this result are stated inside the Description, and the configuration is given in Status.*

## 4 Implications

Not recorded as a separate field for this run. What the disposition means for a user of the core follows from the Disposition meaning in Status and the bounds in Limitations and scope.

## 5 Action

### Checker used

checker\_333.sv

## 6 Evidence

| Ref | Type           | Path, relative to the run directory       | File     | Result recorded                                                                                                               |
|-----|----------------|-------------------------------------------|----------|-------------------------------------------------------------------------------------------------------------------------------|
| E1  | counterexample | formal/prefix_bmc5/job/engine_0/trace.vcd | not kept | pre-fix: csr_illegal_o=1 in EX (cycle 6), instruction retires in WB (cycle 7) without exception; cex_prefix_dbgcsr.png        |
| E2  | proof          | formal/master_prove5/sby.log              | present  | master: abc pdr unbounded PASS                                                                                                |
| E3  | proof          | formal/master_cover4/sby.log              | present  | master: covers hit@7, hit_traps@7 (exception taken), dbg_ok@8 (legal in debug mode)                                           |
| E4  | counterexample | formal/master_bmc4/job/engine_0/trace.vcd | not kept | diagnostic (v4): trigger match on the CSR instruction address -> DEBUG_TAKEN suppresses the exception legitimately; led to v5 |
| E5  | counterexample | formal/master_bmc3/job/engine_0/trace.vcd | not kept | diagnostic (v3): instruction-bus error injected by the solver -> cause INSTR_FAULT; led to quiet-bus assumption               |
| E6  | diff           | cores/cv32e40x_hist (git show a34730a6)   | not kept | fix: ex_stage csr_is_illegal qualifier inverted (X_EXT=0 case). Read after proving.                                           |

Paths are relative to runs/e40x-333. The File column is a check made when this document was generated. It reports whether the cited file exists, not whether its contents were re-validated. 'not kept' means a repository rule deliberately excludes that artefact, usually a waveform dump; rerun the check to regenerate it. 'MISSING' means the file is absent and was expected to be there.

### What each class of artefact establishes

- counterexample. A concrete input sequence produced by the engine. Shows the stated condition is reachable; it is not itself a proof.
- proof. Proof or engine run. Scope depends on the mode: prove is unbounded under the stated model, bmc is bounded to its depth, cover establishes reachability only.
- diff. Source or configuration difference.

Artefact(s) E1, E4, E5, E6 are not retained under repository policy. The result recorded against them stands on the engine log cited alongside; the artefact itself must be regenerated by rerunning the check before it can be inspected.

## 7 Limitations and scope

- The fix is verified against the checks listed under Evidence only. This is not a regression sign-off and does not cover the wider test suite.
- This is an agent investigation record. It requires engineering review before it is used as a disposition. No report generator can supply missing evidence.
- Results apply to the configuration and core revision named in Status. A different configuration needs a separate run.
- Measured tool time is the time the tools ran. It is not an engineering effort figure and not a turnaround commitment.
- Public demonstration runs on open cores are not client engagements and not production verification results.

## 8 Provenance and measurement

**Run directory** runs/e40x-333

**Title source** Title and configuration read back from reports/issues/cv32e40x\_issue\_333.pdf, the

|                  |                                                                                                                                                                                            |
|------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Inputs shown     | previously published report for this issue.                                                                                                                                                |
| Human record     | title, body<br>PR #335 / commit a34730a6 (2021-12-01): ex_stage csr_is_illegal = csr_illegal_i && !(xif_en && xif_accepted); previously && (xif_en && !accepted), always false without XIF |
| Agent confidence | 0.9. This is the investigating agent's own estimate. It is not a measurement and not a reliability figure.                                                                                 |
| Iterations       | 5                                                                                                                                                                                          |
| Run by           | Claude directly                                                                                                                                                                            |

Measured tool and step time

| Step                                                      | Measured         |
|-----------------------------------------------------------|------------------|
| read issue + pre-fix checkout + wrapper                   | 7.0 min (420 s)  |
| v1 checker: vacuous (csr_en)                              | 5.0 min (300 s)  |
| v2 instruction-word: pre-fix CEX                          | 4.0 min (240 s)  |
| v3 quiet bus (bus-error artifact)                         | 5.0 min (300 s)  |
| v4 wb_valid + ns DEBUG_TAKEN (trigger artifact)           | 5.0 min (300 s)  |
| v5 cs DEBUG_TAKEN + trigger: pre-fix CEX, master pdr PASS | 4.0 min (240 s)  |
| compare fix a34730a6                                      | 1.5 min (90 s)   |
| report                                                    | 60.0 s           |
| Total                                                     | 10.9 min (653 s) |

Times are wall-clock for the step named. Some steps may have run concurrently, so they do not sum to an isolated performance figure, and they are not an engineering effort measure.

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